

WASP-52b

R-1028

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_161013 · created 2026-07-30T06:02:53+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457672.9478 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.061
Transit depth (Rp/Rs)^2	2.91 +/- 2.09 [%]
Orbital Inclination (inc)	81.52 +/- 2.73
Airmass coefficient 1 (a1)	0.44 +/- 0.23
Airmass coefficient 2 (a2)	0.59 +/- 1.05
Scatter in the residuals of the lightcurve fit is	49.97 %
Best Comparison Star	#1 - [116, 480]
Optimal Aperture	11.68
Optimal Annulus	24.34
Transit Duration (day)	0.0 +/- 0.0

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.171 $\pm$ 0.061 vs 0.1646 (deviation 0.1 σ)
Duration fit vs published	0.0 d vs 0.0758 d (deviation inf σ)
Mid-time O–C	-2518.1 min
Depth SNR	2.8
Residual scatter	49.97 %

Light curve

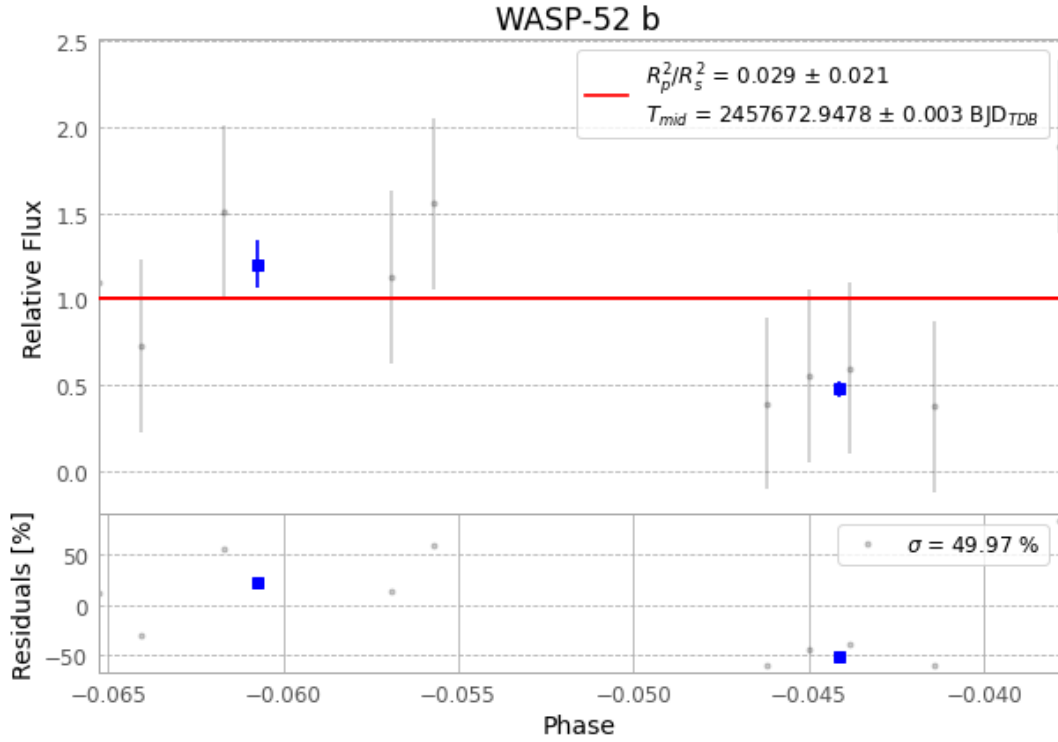


Figure 1 — FinalLightCurve_WASP-52 b_2016-10-12.png (SHA-256 029364394a168c01...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [256, 119], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 472cf55e07e47846...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit d840746

Data provenance

100 FITS frames (66 MB), WASP-52161013014812.FITS → WASP-52161013064512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-161013.log (924c531f47ab...), FinalLightCurve_WASP-52 b_2016-10-12.png (029364394a16...), FinalLightCurve_WASP-52 b_2016-10-12.csv (4a01421a061c...)

Compute resources

Wall time 170.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 06:02Z.